Streamlining Data Mobilization: An end-to-end data analysis demonstrator from a data repository to cBioPortal via Galaxy

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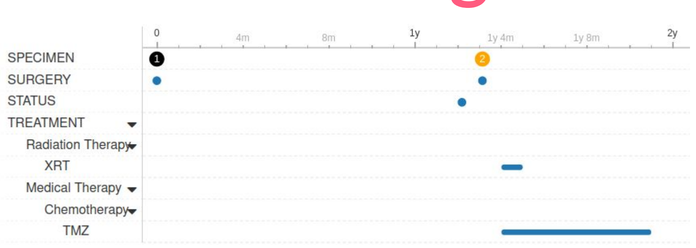
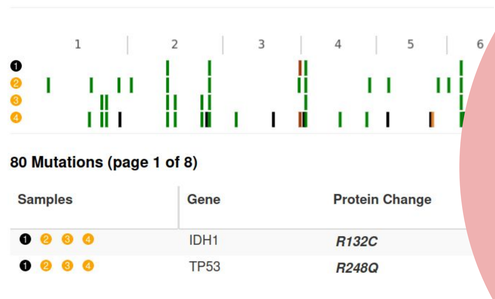
Often, research data is available through repositories and archives with varying access mechanisms depending on the nature of the data. However, it is not usual for such repositories to have the capabilities to process, integrate, and visualise a particular dataset of interest. To facilitate these operations, a framework is necessary for reproducible results and interoperability. We present in this work a technical demonstrator showcasing the mobilisation of data across platforms, illustrating a workflow from a data repository, from NextCloud to cBioPortal via Galaxy with synthetic clinical and genomic data generated in the context of EOSC4Cancer.

Longitudinal Data





4. Full Workflow

In this QR there is a video explaining the whole process, from raw data in a repository, to visualization of the results in cBioPortal.

In the near future...

Data will be extracted to Galaxy directly from EGA

3. Direct Upload to cBioPortal

The MAF files (genomic data) obtained from Galaxy and the synthetic clinical data extracted from the OMOP CDM can be uploaded to cBioPortal to create a new study.

2. Mobilisation of data from repository to Galaxy

Demonstrator where a Galaxy instance in Germany takes raw sequencing data from a Nextcloud Server at BSC (Spain), and runs two workflows to identify somatic variants and convert the results to cBioPortal format.

Workflow (WF) 1: Raw data (FASTA)→Alignment (BAM)→Variant Calling (VCF)

WF 2: Identified Variants (VCF)→Annotation (VCF)→Conversion to cBioPortal (MAF)

1. Creating Synthetic data

Simulation of longitudinal data to represent the cancer journey of four patients with **EHR information** (treatments, relapses and identified mutations) and **genomic data** (normal and tumor samples harboring several subclones that define the evolution of the tumor) along the course of the disease.

OMOP CDM

- Created with PySynth[1]
- Based on Mutations from PCAWG dataset
- Statistics of real CRC patients in Spanish Cancer Unit

Genomic Data

- Created with NEAT GenReads[2]
- Reads simulated from each clone individually and by mixing clonal proportions
- Paired end FASTQ file format

[1] https://gitlab.bsc.es/health-data/pysynth
[2] https://github.com/zstephens/neat-genreads

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